How ADHD Programmers Use Reddit to Elicit Software Requirements and Approach Design

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***Abstract*—Abundant software technologies are marketed as helping users with Attention Deficit Hyperactivity Disorder (ADHD) but often fail to deliver meaningful impact to their intended users. At the same time, along with the popularity of social media platforms and forums, the online community acts as a valuable venue for marginalized populations to perform collective sense-making and challenge the status quo. Our study looks into how ADHD programmers use online forums to build communities, discuss software and better learn their community. By qualitatively analyzing the posts and comments related to software technology, we find that ADHD programmers from diverse backgrounds actively exchange their experience and expertise while seeking advice on building effective software tools for users with ADHD. They also propose novel ways to navigate existing software solutions to meet their own goals. When discussing software, ADHD programmers deliberately advocate for more accessible design and question the norms that negatively impact the vulnerable population. Our study sheds light on this sparsely investigated collective made by ADHD programmers to inform future approaches for more usable and accessible design.**

***Index Terms*—Neurodiversity, Attention Deficit Hyperactivity Disorder, Requirements for ADHD Users.**

I. Introduction

As one of the neurodivergent conditions, Attention Deficit Hyperactivity Disorder (ADHD) has attracted increasing attention due to its broad impact [1]. As with other software technology users, people with ADHD use software tools to instrument various aspects of their personal and professional lives. The market for assistive software tailored to users with ADHD has grown substantially. For instance, when searching 'ADHD' or its related terms on Google Play, Apple Store and eight digital libraries for academic research, Hernandez et al. [2] counted 3,474 software released between 2013-2023. Those software serve a variety of functions for users with ADHD, covering organization, time management, social skills, cognitive training and more. However, several authors question the practical efficacy of those software products and their impact on the intended population [3].

Designing software for users with ADHD requires expertise in both software design and a deep understanding of the user group, their values, goals, and challenges, to contextualize the explicit and implicit impacts of the software on those users [4]. Requirements and guidelines for neurodivergent users (including users with ADHD) have been proposed, e.g., through surveying the prior literature [5] and studying the population directly [6]. However, adopting and interpreting those guidelines and standards still requires extensive experience of designing for those users and balancing competing concerns [5].

In this work, we draw attention to programmers with ADHD, who are on the frontier of building software and, at the same time, have first-hand experience of ADHD. This population is uniquely positioned to push for assistive and neurodivergent-inclusive software. However, this potential is not examined in prior work (e.g., from academic research [7] and grey literature [8]), which mainly focuses on how to help programmers with ADHD navigate the organizational and cognitive challenges associated with their ADHD symptoms.

The goal of this study is to examine how online spaces provide a fertile ground to gather software requirements of users with ADHD, and to emphasize the vital role that programmers with ADHD play in creating effective software for themselves and other users with ADHD. Guided by this overall objective, we adopted a qualitative approach [9] to collect and analyze 127 posts and 751 comments about programmers and software tools from online ADHD forums on Reddit (*r/ADHD*, *r/adhdwomen*, and *r/ADHD_Programmers*). We identified two themes grounded in those online discussions. The first theme concerns how the discussion of software happens among programmers with different backgrounds, from indie to veteran programmers, why they create the posts, and the community's responses. The second theme centres around gathering and contextualizing the software requirements of users with ADHD. ADHD forums provide a space for software developers with ADHD to better understand what users need in their ADHD applications, and advocate for simple, customizable applications that can keep them motivated in their everyday activities.

Our results suggest that people with programming backgrounds and ADHD use online forums to seek collaboration from their peers—collectively brainstorm, implement, and test software. They engage in this community through learning from each other's experiences and expertise, writing solutions to improve existing tools or create new tools, and sharing knowledge to improve the usability and accessibility of the software. They also criticize the existing practices that have dominated the development of software products and their business models. Complementary to existing work of partic-

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ipatory approaches, our work highlights the potential of a different pathway to achieve more inclusive design, a bottom-up approach led by neurodivergent programmers. It calls for future research to create sites that cultivate user-led innovation and investigate platforms and tools that can lower the barriers for people with ADHD to design and build software.

II. Related Work

In this section, we summarize how our work is built on previous literature on experiences of ADHD professionals in software engineering settings and requirement elicitation practices on online forums.

A. Neurodiversity in Software Engineering Setting

Several previous work investigated challenges faced by neurodivergent technology professionals during their training and practice. Morris et al. [10], Gama et al. [11] and Liebel et al. [12] discussed struggles in software development and workplace communications, such as difficulties in organizing and planning tasks, estimating deadlines, task initiation and completion, and even performing mundane activities such as writing boilerplate code. With the shift to work-from-home practices during the pandemic, Das et al. [13] drew attention to difficulties experienced by professionals with ADHD, in using video conferencing software, such as virtual distractions, sensory stimulations and expressing non-verbal communications. Newman et al. [7] analyzed the *r/ADHD_Programmers* subreddit, identifying that programmers experience challenges involving cognition, time, distractions, communication, and alignment. Programmers with ADHD adopt different strategies, such as adjusted training material, body doubling [14], medication, pomodoro timers [15], use of assistive technology, structured feedback, and written communication or even seeking better aligned jobs.

These studies shed light on the day-to-day experiences of neurodivergent software developers and are mainly centred around the workplace environment and interpersonal communication. However, they did not specifically examine potential challenges experienced by employees in using task-management and productivity tools. Complementary to previous work, our study brings attention to how programmers with ADHD navigate their daily use of software and what they consider important when building software for personal projects. We focus on programmers with ADHD, considering that their background in software design and development enables them to challenge and tinker with the space of how existing software is designed.

B. Requirement Elicitation using Online Platforms

Online platforms create a public space for software users to share their experiences and feedback, providing a playground rich for CrowdRE. Pagano and Maalej analyzed randomly selected 1,100 reviews from Apple's AppStore for 22 categories of applications, identifying that approximately 33% of the feedback can be immediately beneficial to requirement analysts [16]. Information mined from online spaces can include useful information to application developers, such as bug reports, experience summaries, new feature ideas, performance ratings etc. Several authors have successfully mined useful data that can be refined into requirements artifacts from platforms like Reddit [17], Twitter [18], YouTube [19], TikTok [20], and application stores like Google Play, Google+, Apple App Store, Windows Phone Store [21]. Tizard et al. [22] suggest that online product forums are robust sources of information, where users often provide detailed contexts of encountered software problems, and other users respond with confirmations or workarounds to the problem.

In summation, prior literature thoroughly presents promising ventures in using online forums for requirement elicitation, even presenting tools to mine the relevant information [21], [23]. Our study further explores the use of online forums for understanding the needs of vulnerable populations, particularly that of users with ADHD, who rely on software to manage their unique and challenging symptoms.

III. Method

This section provides details of our study design and the considerations that drive several decisions when conducting the study and reporting its results.

A. Data Collection

We began by examining public forums for users with ADHD, with a substantial user base, either those mentioned in the previous work or those returned by querying 'ADHD forums' on search engines. From an initial list of seven public forums, we excluded two forums that were dedicated to children and teens with ADHD. We then searched the remaining forums to identify if they contained topics relevant to our research objective using terms 'ADHD tools', 'ADHD software', 'ADHD engineer', 'ADHD developer'. The first author then read through a subset of retrieved posts to identify their fitness. After this round of initial exploration, we eliminated two more forums due to the limited presence of programmers and developers with ADHD. The remaining three forums are all hosted on Reddit, i.e., *r/ADHD_Programmers*, *r/ADHD*, and *r/adhdwomen*. Among them, *r/ADHD_Programmers* is dedicated to programmers and software professionals with ADHD; *r/ADHD* is a peer support group for people with ADHD; *r/adhdwomen* is a space for women with ADHD to find support and discuss their experiences.

We read through rules on the three forums to ensure that they do not forbid or discourage research use of the forum data. We then performed another round of querying and filtering on posts and comments between 1st January 2023 and 31st December 2023 on those forums. Our search queries were designed to include variations of reference to modern software, including apps, tools, websites and APIs. We also included different terms referring to individuals with programming backgrounds, i.e., developer, programmer, and engineer. Therefore, our final search queries were *(programmer AND (api OR tool OR app OR website)), (developer AND (api OR tool OR app OR website)) and (engineer AND (api OR tool*

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OR app OR website)). We extracted a total of 474 posts across the three forums that matched our queries and then performed manual filtering. The filtering ensures that in-scope posts and comments were written by and for software programmers, who discuss the design, development, or feedback about tools to help them manage symptoms of ADHD. Any posts and comments that were marked as deleted, redacted or retracted by the platforms, along with the replies following them were excluded from our analysis. Our final dataset consisted of 127 posts and 751 comments on these posts.

B. Data Analysis

The forum posts and comments were analyzed qualitatively and iteratively. We follow an inductive thematic analysis process [9]. During the manual querying and filtering, we started getting familiar with the data, including the overall topics and concentrations of the discussion. When we started coding the data, we first randomly sampled 15 posts and their corresponding discussion threads in the comments and individually open-coded all the content that was relevant to our research objective. All authors met and discussed their codes, and revised the codes to formulate the central themes from the data. We reflected on multiple views of the data encouraged by our research objective, such as why forum users want to contribute to discussions online, the role of software in managing their ADHD symptoms and the rationale behind developing their own tools for managing ADHD. Our discussion also brought up our code of software features that developers with ADHD considered useful, their perception of tools and the design considerations. Based on the discussion, we developed an initial codebook. Here, the codebook was not meant to ensure coding reliability, but it acted as a living document to capture details of the researchers' reflexive process and scaffold the coding. The first and second authors continued to iteratively code another 20 posts of the dataset. Any updated thoughts and disagreements on the codes were discussed in detail with all authors of this work. The first author then coded the remaining data using inferences and understanding derived and documented from previous rounds of coding.

IV. Results: Requirement Elicitation on Reddit

We present our main findings in two parts. In this section, we first describe how ADHD subreddits are used as a platform to connect programmers with ADHD with other software users with ADHD to better understand the needs of their community. We describe how conversations of requirement gathering occurs on an online platform and how forum members respond to them. In Section V, we describe our findings on the requirements of ADHD software users, that are discussed on the ADHD forums.

A. Diverse Background of Community Members

The posts from the forums reveal that the forum members possess **various backgrounds and experiences in software development**. We observe many forum members are novice and indie programmers who are either at the beginning of their career path or working on small-scale projects. We also see veteran programmers, e.g., who have been *"retired after a 29-year career."* Their expertise collectively covers the whole stack of software development and product management. For example, one forum member described their occupation and passion for usability as *"as a frontend developer, my interfaces are always of the highest level of quality. I can spend hours trying, testing, and fixing things just to make sure that this path is clear and easy to understand for end users. I'm super mad when people don't pay attention to UI, or maybe more, to general usability and UX of things they create. I also pay great attention to developer experience in libraries."*

At the same time, forum members also demonstrate **varying levels of disclosure about their ADHD symptoms**. In many cases, explicitly disclose their ADHD and describe their personal stories and challenges to contextualize their posts or comments. Their level of disclosure is different, from a brief mention (e.g., *"I recently realized that I have ADHD and have been working on a tool that addresses some of the challenges I face"*) to a description of their formal diagnosis (e.g., *"I just recently got formal diagnosis with an emphasis on execution dysfunction and emotional dis-regulation"*). At other times, they just describe their personal stories and imply their ADHD in other ways, as in *"The thing is, I know that if I commit to something, I can succeed at anything I pick up, but as you know, consistency is hard. I keep getting thoughts about how the current option is not good enough and the others are better, even if I had completely opposite thoughts the day before."* Given the public nature of the forums, we also observe cases when the posts are written by people without ADHD. In those cases, they explicitly describe the purpose of their post, in particular why they want to reach out to this community. Not only do programmers with ADHD discuss their needs and values in their professional life, they also actively contribute their opinions or experience either during using or designing those tools. Those discussions reflect important considerations, novel properties and mechanisms on *how* to meet the need of users with ADHD across multiple dimensions.

B. Posts for Requirement Elicitation from Forum Members

Due to the public nature of the community, members can be programmers with ADHD, software users with ADHD, or both. As developers, they consider the forum as an opportunity for them **to understand their user groups better** to improve the existing features or suggest new features to work on. Often times, their personal experiences and understanding of the current technology landscape motivate them to build novel tools to cater to their needs, as explained in one forum post: *"I have come to the conclusion that the only option is for me to develop my own app."* Since they started developing a software to solve their own struggles, in many cases, they **want to see if their peers also share the same experience** and would benefit by their self-developed tool. Therefore, they create posts to describe their work or ideas and seek immediate feedback on them. *"As a software engineer having all symptoms of a mild*

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form of ADHD - I always needed to write things down. But I've struggled to make any note-taking app work for me. I just forget where I wrote this note, was it in my notes, email or somewhere in work chat... I've been working really hard building this chatbot-based memory assistant for quite some time. [...] What would be your objections to using something like that?"

Where on one hand, some developers post about their ideas for potential ADHD applications, some others directly share their self-designed and developed ADHD tool for other forum members to use. They create these posts to **seek feedback on their applications** from their community, *"I'm still learning what works and what doesn't. Your feedback is critical to helping me make the app even better."* Programmers often share online links to their application, screenshots, or videos of their finished application in their post. Forum members make use of the attached media to provide directed and meaningful feedback to the developer. For instance, when one forum member shares a link to their completed journal application, another forum member comments, *"Just had a quick play on my phone. Seems quite good actually but maybe have a little tour of the application after the user signs in."*

As software developers who struggle with ADHD symptoms, some community members create posts to ask for **suggestions to manage their symptoms and the impact to their work and life**. Such posts often start with a more detailed description of their concrete struggles and end with what they hope to know, either general advise (e.g., *"Any tricks, advices, apps or techniques to stay consistent with your tasks? How do you manage to do this?"*) or concrete software or workflow that can be adopted (e.g., *"Does anyone have a pipeline for integrating pen input across different devices, like a pen display on Windows and an Android tablet?"*). Forum members who self-develop ADHD application often comment on solution seeking posts like these to promote their tool and gather feedback from other community members. One post requesting solutions for time blindness garnered the attention of a few application developers, *"I made this app because I was in your exact situation. It's free. Download it and let your calendar wake you up instead of you fighting to remember everything. Its still a new app, so I'm also looking for feedback!"*

C. Reaction and Attitude to Requirement Elicitation Posts

In most cases, forum members **actively engage in discussion** of their own technology choices and offer their support for the posts that ask for feedback on the software. Sometimes, their support can be seen in their thoughtful comments after thorough and careful user testing. For example, after testing a medication tracking app for a week, one forum member responded with a comment of more than four hundred words. In it, not only did they describe the critical features that are missing (e.g., *"there is no reminder to take your pill."* and *"there is no way to edit the time I took the medicine."*), but they also praised on the aspects that the software did well (e.g., *"visual looks great"*, *"it's easy to navigate"*, etc.) and offered concrete suggestions of other features and considerations (e.g., on visual icons and timing of push notification).

In other times, such support is reflected in their comments **expressing interest to use the software, share it, or collaborate on the software project**, as in *"This looks really promising! I've installed it on my PC and Android phone and will test focus mode when the work week starts, but I can definitely see myself using it after the trial!"* In another example, when a post was created asking for a new Android app that uses gamified tasks, one community member responded, *"I am a web developer. Maybe I could create an app? Who wants to join?"*, followed by another comment by the original poster, *"Yes, I am open to partnerships to create the app."*

At the same time, some forum members are overwhelmed by the sheer volume of requests for software feedback and occasionally express their **skepticism and frustration** as in *"I joined [forum name] thinking it was going to be as sweaty as the [other forum] was in the day [...] Every day, there is some person who wants to make an ADHD miracle tech cure. Does anyone in this [forum name] actually have ADHD?"* and in *"Yet another focus app from a new [forum name] user. I feel like there are at least a couple every month, and they are all completely and utterly useless for me."* In other cases, such skepticism is expressed toward certain technologies, such as AI. In this post, one forum member responded to an AI-based productivity software sarcastically, *"Wow, AI is surely the right way to treat ADHD. I wholeheartedly believe there is no way this can go wrong and that we haven't seen it happen 1000 times before."*

V. Results: Concrete Features and Designs

Applications for ADHD usually follow the common theme, i.e., providing a mechanism for a person with ADHD to organize their daily life. These include tools for task management, time management, distraction blocking, reminders and calendars, habit tracking etc. In this section, we discuss the characteristic features and design considerations that are discussed on the ADHD subreddits.

A. *Desired Features*

In this subsection, we present our analysis of the features of ADHD software that are deemed useful and practical to users with ADHD. These features are useful to keep the themselves accountable as professionals to complete their tasks, manage distractions and challenges with impulsivity and attention control, and stay motivated in their tasks.

*1) **Productivity Tools and Tracking Features**:* To help themselves stay organized and productive in both work and personal lives, individuals with ADHD rely on productivity tools such as reminders and timers *"to keep track of deadlines, visualize how much time is left,"* checklists and task schedulers *"to make sure you get the task done,"* task decomposers to *"break down a task into manageable steps, simplify a complex process or question,"* and so on. Forum members also utilize features to track their progress on tasks, spending habits, medication intake and symptoms etc.

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*2) **Flexible Application Usage Restrictions**:* When combating with distractions, forum members emphasized their need for flexibility for the extent of blocking in distraction blockers. One forum member describe what they need as *"the opposite of all the current 'stay focused' apps."* They elaborated further of what design is actually needed that offers flexible blocking, *"my code is compiling which will take 30 sec to 2 minutes, so I open the browser. I don't want anything to block the browser during that time. But, if I spend more than 5 minutes on the browser, I want a notification to tell me I spent too much so I can go back to work. Ideally the timer should reset when the browser is out of focus."* Moreover, a forum member mentioned the need for certain amount of distraction because of their calming effect. They even hope to use some "fun" apps throughout the day. Therefore, what they need is a productivity app that *"from 9-5, Monday-Friday, I can use my 'fun apps' for a total of 10 minutes each hour and then I'm shut out until the next hour."*

*3) **Integrating AI and Automation**:* Forum members introduce AI products for various purposes, such as task initiation, automation, body doubling etc. When using artificial intelligence, however, forum members prefer human-like AI chatbots, to *"just chat with you like chatGPT but more warm, supportive and human-like. It also has functions like 'Help me plan', 'I just wanna vent' which seem relevant."* When tasked with assignments on writing, members applaud AI's capabilities to provide prompts for writing tasks, because *"looking at a blank paper on your screen automatically triggers feelings of dread"*. With AI writing the basics of the document, *"it's almost always easier to edit something instead of creating from scratch."*

*4) **Customizing Presentation**:* The ability to change visual or audio elements in the software, such as colour, brightness, and sound, can improve the application usage experience for the user. For instance, one forum member shares their experience on choosing the right task management application for themselves, *"Give us the chance to adjust appearance and visual effects (and include some easy default themes to choose from). Sometimes it's a matter of one colour or pattern making all the difference for me to sit and use it."*

*5) **Gamification**:* Task management and time management applications are usually not sufficient for individuals with ADHD to stay motivated to complete their tasks and maintain their everyday routines. It can still be overwhelming and difficult to initiate tasks. One user shares applications to 'gamify' their life because *"turning my tasks into a game has made them less daunting and more enjoyable"*. Another forum member described how they leverage their gamified system, *"every time I complete a habit/daily routine/task, I earn coins. I also set up some reward that cost coins, such like 'play video games for 25min, which costs 20 coin'. [...] The coins is limited, and are earned through my effort in completing habit/daily/tasks."* Another user tied their productivity rewards to govern their personal spending allowance -*"doing tasks in reward myself with a fake golden coin or a few. These represent spending money."* Gamified productivity tools can also include features to *"socialize with your friends, compete on a leaderboard, and earn rewards for reaching certain milestones,"* to leverage supportive communities that *"feel like you're part of a team all working towards your goals together."*

*6) **Randomization**:* Forum members propose to inject randomness routines for managing challenges with executive function. Initiating or hustling through a *"wall of awful tasks that you're not quite sure how to start"* can be overwhelming and difficult to tackle. For instance, one forum member creating their own productivity app adds a feature that picks a random task at a given time *"for those moments when you can't decide what to tackle."* In dealing with time blindness, a challenging symptom when an individual with ADHD may not be able to comprehend the passage of time, one forum member suggested creating an application that adds randomness into time management to force its users to stop and look at the clock. The author explained the feature and rationale as *"I've tried setting my office clock a few minutes fast, but now I know it's a few minutes past, and it doesn't work anymore. I'm still late for meetings. If I had a clock that would randomly change time overnight to either a few minutes fast or a few minutes slow, I would have to stop and look at the clock and then check my computer clock. I think this would be great for those with time blindness issues."*

B. Design Considerations

When designing self-developed ADHD applications or providing feedback to others, forum members bring up important considerations that have crucial implications for designing and deploying software for users with ADHD. In this subsection, we present the three important design ideas discussed on ADHD forums.

*1) **Ease of Use**:* Avoiding overwhelming users visually and functionally is highly valued by forum members in terms of software design. It includes, easy navigation through the application, easy access to application features, sync across different devices and other elements that can make an application easy to use. When providing feedback to a productivity app, one forum member explicitly said that they prefer *"having it be a simple interface / not overly complicated"* and *"apps that are too complicated to figure out, I will give up on."*

*2) **Privacy and Security**:* Programmers with ADHD suggest the need to provide control on how personal data is accessed and stored by the software. For instance, when a user presented an application to automatically track purchases and bill payments through banking APIs, another forum member asked if transactions can be inputted manually instead: *"while I agree that manual data entry can be a bit of a chore, it is one I am willing to accept because I prefer to avoid granting 3rd party access to my banking information."* They continued their concern over the application's data policy, questioning the developer over *"where is the user's data saved? how securely is it stored? is there an option for offline storage only? Ideally I would like to save as an offline file with encryption."*

Another instance of data wariness occurs when applications require users to register or sign-up. In one post introducing

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an application for overcoming stressful feelings and task procrastination, when one user comments *"sign-up being required feels like you're gathering data on everyone's darkest fears"*, another user suggests the developer to permit guest users – *"you can attract more non-account users! even though not 100% of users will have accounts, the portion that do might still be a higher number overall"*. At other times, members question the need for personal data required when a new user registers to an application, *"why do you ask for birthday? I don't feel like you need that info for anything at all."*

3) ***User Engagement****:* One common criticism that many forum members have about productivity tools is that they often get bored while using it for long periods of time. Visual features can be used to provide novelty to application users to keep them motivated to use the application. For instance, one user suggested a feature to rotate productivity systems because they *"find a system that actually works and makes me happy, but I can't stick with it because I get bored."* Such an application can keep all the data as usual, but simply change how the system is presented (like Kanban, Gannt, Timeblocking etc.)

Forum members also discussed that the design should draw attention of users with ADHD towards certain components of an application based on the context of the software usage and what is considered important by the users. For instance, some members use progress tracking features to *"keep track of what has been completed and what remains to be done"*. Rather than a plain list of the to-dos, they recommend to *"visualize how much time is left until deadlines"* so that they can stay motivated and *"feel a sense of accomplishment as you complete each task."* One user recommended an application which can offer *"a visual representation of your day with their daily calender view"*, along with colorful tags for tasks *"so each task has a different color depending on what it is (errand, at home, focused work etc.)"* Similarly, another forum member finds such visual representations *"make your schedules and to-do lists more comprehensible."*

VI. Discussion and Implications

ADHD is a complex condition with a broad spectrum of symptoms that affect each individual differently. Designing a software to meet the requirements of individuals with ADHD is an obscure task. On one hand, traditional requirement engineering practices such as interviews, workshops, focus groups, surveys etc., are time consuming and limited in their capabilities to gather requirements from a broad spectrum of participants [24]. These methods also do not take into consideration the evolving requirements of a population. On the other hand, it can also be challenging to accommodate the communication preferences, sensory distractions and stressors for a participant with ADHD, when working with traditional RE practices [25].

In contrast, CrowdRE methods assemble collective insights and experience of a diverse community to derive validated user requirements. Online forums like Reddit become a fertile soil for CrowdRE innovation because *push feedback* (a process in CrowdRE when the crowd shares their feedback without being prompted [26]) on ADHD software is frequently, spontaneously and indigenously generated via user posts. Furthermore, Reddit provides a space for users to communicate asynchronously and non-verbally, according to their comfort levels of engagement [27], thus becoming a comfortable space for individuals with ADHD to discuss their interests, frustrations and needs about various issues, including ADHD software.

We find that requirements elicited on Reddit can adequately echo the concepts identified in prior literature. For instance, our study reproduces some design considerations for software tools identified in previous work such as customizability [28], [29], ease of use [30], [31] and adoption of AI [32]. Requirement elicitation and gathering on Reddit also provide new, rich and personalized insights into the unique needs of ADHD users innovative new functional (like gamification, randomization and access control) and non-functional requirements (like user engagement, and privacy and security). It indicates the relevance of the forums for ADHD programmers to get engaged with their community and to design more inclusive and usable software for ADHD users.

For CrowdRE researchers and practitioners, our findings suggest the value in exploring public forums, which provide an open and non-clinical space to understand the sentiment of our participants. We propose that mining user feedback and opinions using online forums such as Reddit can be both economical and effective for vulnerable user groups. We show that online forums can not only reproduce the findings from traditional research techniques like interview studies, workshops, and lab studies, but also provide richer and longitudinal information from a broader demographic. With a nod to the reliability of Reddit and other online spaces in our journey to CrowdRE Ten, we propose that unstructured data from online forums can be processed into useful requirement artifacts with proper tools and automation, a goal for CrowdRE TwentyThirty.

VII. Conclusion

In this work, we examine the online community of ADHD programmers and their potential in CrowdRE. Our analysis finds that programmers with ADHD use online forums to explore software features to implement, exchange knowledge and expertise, and seek feedback and advice to improve self-developed tools. CrowdRE practices can avoid a one-size-fits-all mindset and derive the challenges and evolving needs of ADHD users. We find that existing software often fails to meet the personal goals of programmers with ADHD, such as being accountable and staying motivated. Meanwhile, ADHD users look for simple but effective features and customizable software that minimizes the risks on users from vulnerable populations. Our work contributes to the existing literature on recognizing and prioritizing the living experience of users with ADHD when designing software for them and re-iterates reliability of Reddit as a rich forum for user feedback suitable for CrowdRE.

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REFERENCES

[1] P. Song, M. Zha, Q. Yang, Y. Zhang, X. Li, and I. Rudan, "The prevalence of adult attention-deficit hyperactivity disorder: A global systematic review and meta-analysis," *Journal of global health*, 2021.

[2] J. Hernández-Capistrán, G. Alor-Hernández, L. N. Sánchez-Morales, and I. M.-C. and, "A decade of apps for adhd management: A scoping review," *Behaviour & Information Technology*, 2025.

[3] H. Lindstedt and Õ. Umb-Carlsson, "Cognitive assistive technology and professional support in everyday life for adults with adhd," *Disability and Rehabilitation: Assistive Technology*, 2013.

[4] K. Spiel, E. Hornecker, R. M. Williams, and J. Good, "ADHD and technology research - investigated by neurodivergent readers," in *CHI '22: CHI Conference on Human Factors in Computing Systems, New Orleans, LA, USA*.

[5] I. Rocha, A. Moreira, J. Araújo, and G. Liebel, "Envisioning a requirements elicitation method for neurodivergent-inclusive software," in *2025 IEEE 33rd International Requirements Engineering Conference (RE)*.

[6] C. Putnam and L. Chong, "Software and technologies designed for people with autism: What do users want?" in *Proceedings of the 10th International ACM SIGACCESS Conference on Computers and Accessibility*.

[7] K. Newman, S. Snay, M. Endres, M. Parikh, and A. Begel, "'get me in the groove': A mixed methods study on supporting adhd professional programmers," in *2025 IEEE/ACM 47th International Conference on Software Engineering (ICSE)*. IEEE Computer Society.

[8] D. Fernandez. (2025) How adhd affects me as a programmer and how i navigate it.

[9] V. Braun and V. Clarke, *Thematic Analysis: A Practical Guide*, 2011.

[10] M. R. Morris, A. Begel, and B. Wiedermann, "Understanding the challenges faced by neurodiverse software engineering employees: Towards a more inclusive and productive technical workforce," in *Proceedings of the 17th International ACM SIGACCESS Conference on Computers & Accessibility, ASSETS 2015, Lisbon, Portugal*.

[11] K. Gama, G. Liebel, M. Goulão, A. Lacerda, and C. Lacerda, "A socio-technical grounded theory on the effect of cognitive dysfunctions in the performance of software developers with adhd and autism," in *2025 IEEE/ACM 47th International Conference on Software Engineering: Software Engineering in Society (ICSE-SEIS)*.

[12] G. Liebel, N. Langlois, and K. Gama, "Challenges, strengths, and strategies of software engineers with ADHD: A case study," in *Proceedings of the 46th International Conference on Software Engineering: Software Engineering in Society, ICSE-SEIS2024, Lisbon, Portugal*.

[13] M. Das, J. Tang, K. E. Ringland, and A. M. Piper, "Towards accessible remote work: Understanding work-from-home practices of neurodivergent professionals," *Proc. ACM Hum. Comput. Interact.*, 2021.

[14] T. Eagle, L. B. Baltaxe-Admony, and K. E. Ringland, "Proposing body doubling as a continuum of space/time and mutuality: An investigation with neurodivergent participants," in *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility, ASSETS 2023, New York, NY, USA*.

[15] F. Cirillo, *The Pomodoro Technique: The Acclaimed Time-Management System That Has Transformed How We Work*. Crown Currency, 2018.

[16] D. Pagano and W. Maalej, "User feedback in the appstore: An empirical study," in *2013 21st IEEE international requirements engineering conference (RE)*.

[17] T. Iqbal, M. Khan, K. Taveter, and N. Seyff, "Mining reddit as a new source for software requirements," in *2021 IEEE 29th international requirements engineering conference (RE)*.

[18] G. Williams and A. Mahmoud, "Mining twitter feeds for software user requirements," in *2017 IEEE 25th International Requirements Engineering Conference (RE)*.

[19] O. Karras, E. Kristo, and J. Klünder, "The potential of using vision videos for crowdre: Video comments as a source of feedback," in *2021 IEEE 29th international requirements engineering conference workshops (REW)*.

[20] M. Sihag, Z. S. Li, A. Dash, N. N. Arony, K. Devathasan, N. Ernst, A. B. Albu, and D. Damian, "A Data-Driven Approach for Finding Requirements Relevant Feedback from TikTok and YouTube," in *2023 IEEE 31st International Requirements Engineering Conference (RE)*.

[21] A. Di Sorbo, S. Panichella, C. V. Alexandru, J. Shimagaki, C. A. Visaggio, G. Canfora, and H. C. Gall, "What would users change in my app? summarizing app reviews for recommending software changes," in *Proceedings of the 2016 24th ACM SIGSOFT international symposium on foundations of software engineering*.

[22] J. Tizard, H. Wang, L. Yohannes, and K. Blincoe, "Can a conversation paint a picture? mining requirements in software forums," in *2019 IEEE 27th International Requirements Engineering Conference (RE)*.

[23] M. Nayebi, K. Kuznetsov, A. Zeller, and G. Ruhe, "Recommending and release planning of user-driven functionality deletion for mobile apps."

[24] R. Snijders, F. Dalpiaz, M. Hosseini, A. Shahri, and R. Ali, "Crowd-centric requirements engineering," in *2014 IEEE/ACM 7th International Conference on Utility and Cloud Computing*.

[25] V. G. Motti, "Designing emerging technologies for and with neurodiverse users," in *Proceedings of the 37th ACM International Conference on the Design of Communication, SIGDOC 2019, Portland, OR, USA, October 4-6, 2019*.

[26] E. C. Groen, N. Seyff, R. Ali, F. Dalpiaz, J. Doerr, E. Guzman, M. Hosseini, J. Marco, M. Oriol, A. Perini *et al.*, "The crowd in requirements engineering: The landscape and challenges," *IEEE software*, 2017.

[27] T. Mok, A. Tang, A. McCrimmon, and L. Oehlberg, "Experiences of autistic twitch livestreamers: "i have made easily the most meaningful and impactful relationships"," in *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility, ASSETS 2023, New York, NY, USA*.

[28] H. H. Zhu, R. Chen, and Y. Zhao, "Focusview: Understanding and customizing informational video watching experiences for viewers with adhd," in *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, 2025.

[29] R. Deshmukh, "Toward neurodivergent-aware productivity: A systems andai-based human-in-the-loop framework for adhd-affectedprofessionals," in *Proceedings of the 16th Biannual Conference of the Italian SIGCHI Chapter*, 2025.

[30] L. Jiang, W. Ko, S. Yuan, T. Shende, and S. Azenkot, "Shifting the focus: Exploring video accessibility strategies and challenges for people with adhd," in *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*.

[31] R. Gunawardana, M. Perera, R. Lakshika, C. Ranasinghe, and K. Karunanayaka, "Enhancing adaptive personalized learning interfaces with generative ai for individuals with adhd," in *Proceedings of the 16th International Conference of Human-Computer Interaction (HCI) Design & Research*, 2025.

[32] A. Pinto, E. Quilter, J. Vasikaran, S. Koerner, T. Chimonas, E. E. Nielsen, P. Marshall, and K. O'Hara, "'a little bit of a life raft'–exploring the use and experiences of chatgpt as a support tool among adults with adhd," in *Proceedings of the 37th Australian Conference on Human-Computer Interaction*, 2025.

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